

B.Tech. Degree IV Semester Examination April 2014

ME 1406 MANUFACTURING PROCESS (2012 Scheme)

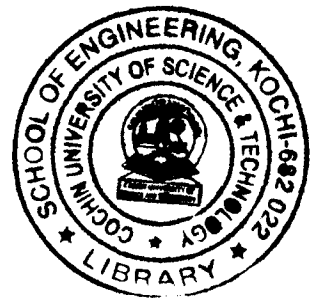
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain advantages and limitations of metal casting process.
- (b) Write notes on (i) gating ratio (ii) risering design
- (c) Describe centrifugal casting.
- (d) Describe die casting.
- (e) Write notes on different types of sheet metal operations.
- (f) Explain different types of forging operations.
- (g) Briefly describe advantages of electrode coating in arc welding.
- (h) Explain thermit welding process.



PART B

(4 × 15 = 60)

- II. Explain pattern allowances, with the help of neat sketches.
- OR**
- III. Explain the preparation of moulding sand. What are the desirable properties of moulding sand?
- IV. Explain shell moulding. Describe the process in detail.
- OR**
- V. Explain any four types of casting defects.
- VI. Narrate the process of direct extrusion. Describe how a hollow profile is extruded, with the help of a neat sketch.
- OR**
- VII. Describe rolling process in detail. What is roll passes?
- VIII. Write notes on different types of resistance welding.
- OR**
- IX. Write notes on
 - (i) Laser beam welding
 - (ii) Diffusion welding
 - (iii) Electro slag welding
